

ICT Skills

1.	An operating system is the basic software that controls the computer. It serves as an interface between the user and the computer. Some of the functions of Operating system are: ✓ It manages all the devices of a computer and keeps track of the status of the device, whether it is busy or not. ✓ It also checks whether the device is functioning properly or not. ✓ It also controls software resources of the computer. ✓ It manages the computer memory and keeps track of which memory space is in use by which program and which space is free. ✓ It manages the structure of the files and directories on a computer system. ✓ It keeps track of the amount of disk space used by a specific file. ✓ It allows you to create, copy, move and delete files.
2.	Types of Operating Systems ✓ DOS DOS (Disk Operating System) ✓ Windows It is an operating system developed by Microsoft. ✓ Linux It is an operating system designed for personal computers. It is a free and open-source software ✓ Mobile operating Systems: Android, Symbian, Windows Phone, iOS
3.	Types of Operating Systems The different types of operating systems are as follows: Interactive (GUI-based) A graphical user interface is a user-friendly operating system in which commands can be entered by clicking/double-clicking/right clicking a mouse. Windows operating system is an example of an interactive operating system. Single-user , single-task operating system This type of operating system allows only one user to do a task on the computer and one thing at a time. Single-user, multi-task operating system This type of operating system is used on Desktop computers, laptops, where a single user can operate on several programs at the same time. For example, Windows, Apple MacOS are examples of single-user multi task operating system. Multi-user A multi-user operating system enables multiple users to work on the same computer at different times or simultaneously. Real Time A real time operating system is used to control machinery, scientific instruments, like robots, in complex animations and computer controlled automated machines. A real-time operating system is a computing environment that reacts to input within a specific period of time. It manages the resources of the computer so that any particular operation is executed in the same amount of time every time it is executed. Windows CE and Lynx OS are examples of real-time operating systems. Distributed A distributed operating system runs on a set of computers that are interconnected by a network. It combines the different computers in the network into a single integrated computer and storage location. Windows, UNIX, and LINUX are examples of distributed operating systems.
4.	Windows 7 is an operating system developed by Microsoft and is used on personal computers. After loading Windows 7, the first screen that appears on the monitor is called desktop. From the desktop, you can access different components of Windows 7. By default, Windows 7 has a picture for the desktop background. This is called wallpaper. Small pictures on the desktop are called icons. These icons represent files, folders, applications, etc. At the bottom of the desktop is a long bar called the Taskbar. To the left of the taskbar is the Start button. Documents: This contains area to store files on the computer Taskbar is the long horizontal bar present at the bottom of the screen. To the left is the Start button. Right of the taskbar contains Date/Time. You can also see icons of active applications and some shortcuts on the Taskbar
5.	Shut Down:- allows the user to turn off the computer All Programs:- Provides access to all the installed programs and applications. To start an application, just click it. Search Box:- This allows the user to search a file or a folder or run executable files. Control panel:- This allows the user to change various settings. Recycle Bin: Files and folders deleted by the user are stored in the Recycle Bin. From Recycle Bin, you can retrieve files or folders deleted by mistake.

6.	1. What is an operating system? 2. Mention any three functions of operating system. 3. Name any three operating systems for computers 4. Name any three mobile operating systems. 5. What is the name given to the screen that appears after Windows operating system is loaded? 6. Small images on the desktop are called _____ 7. Name two special icons on the desktop . 8. Name the icon that provides access to all the drives, files and folders on a computer. 9. Name the button to the left of the taskbar 10. Picture for the desktop background is called _____.
7.	1. Similar types of files can be placed into groups called _____. 2. Mention any one way in which can create a new folder. 3. To change the name of the folder, right-click and select _____ option from the Shortcut menu. 4. How can you delete a folder? 5. What is the difference between moving and copying a folder? 6. Differentiate between file and folder. 7. Name the set of commands to move a folder from one location to another. 8. Name the place where deleted files are placed. 9. How can you empty the recycle Bin?
8.	Importance and need of care and maintenance of computer 1. Keep the computer dust free. 2. Do not eat or drink while working on the computer. Food or drink may spill on the system. 3. To keep the keyboard clean, make sure your hands are clean before using it. 4. CDs and DVDs should be handled carefully so that it does not get any scratches. 5. Keep keyboard covered when not in use.
9.	General precautions to be takes while cleaning the computer components are: ✓ Always Power Off the computer system before cleaning. ✓ Never spray cleaning fluid directly on the component of the computer. First spray the liquid on the cloth and then wipe the component. ✓ Do not allow the cleaning liquid to drip near the circuit board. ✓ Preferably use anti-static wrist band which helps to prevent building up of static electricity near electronic device.
10.	1. What is the possible cause of slowing down of computer? 2. Why keyboard should be covered if not in use? 3. What general precautions should you take while cleaning the computer components/ 4. List some of the maintenance activities for the computer system. 5. Why regular disk defragmentation should be done?
11.	ICT stands for Information and Communication Technology. ICT refers to all the methods, tools, concepts related to storing, recording and sending digital information. ICT skills help us to communicate, run our business and stay connected with our family and friends. Hence, every person needs to acquire ICT skills and build them to stay updated with the latest software and applications (apps).
12.	Basic Functions performed when a computer starts Just as we go about our morning routine without anyone telling us what to do, a computer automatically runs a basic program called BIOS (Basic Input/Output System) as soon as it is switched on or the power button is pushed on. The BIOS first does a self-test. If the self-test shows that the system is fine, the BIOS will load the Operating System.
13.	Keys labeled from F1 to F12 are function keys. Control keys: Keys, such as Control (CTRL), SHIFT, SPACEBAR, ALT, CAPS LOCK and TAB, are special control keys that perform special functions depending on when and where they are used. Enter key: You use the ENTER or the RETURN key to move the cursor to the beginning of a new line. In some programs, it is used to send commands and to confirm a task on a computer. Navigation keys: Keys, such as the arrow keys, HOME, END, PAGE UP, and PAGE DOWN are navigation keys.

	Command keys: Keys, such as INSERT (INS), DELETE (DEL), and BACKSPACE are command keys.								
14.	Some of the shortcut keys: <table> <tr> <td>CTRL+z — undo</td><td>CTRL+c — copy</td></tr> <tr> <td>CTRL+y — redo</td><td>CTRL+v — paste</td></tr> <tr> <td>CTRL+a — select all</td><td>CTRL+p — print</td></tr> <tr> <td>CTRL+x — cut</td><td>CTRL+s — save</td></tr> </table>	CTRL+z — undo	CTRL+c — copy	CTRL+y — redo	CTRL+v — paste	CTRL+a — select all	CTRL+p — print	CTRL+x — cut	CTRL+s — save
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15.	Basic Tips for Taking Care of Devices Keeping a Device Clean Cleaning a device, such as a computer or mobile means to keep the screen, keyboard and mouse clean. Keyboard: First we should not eat anything over a keyboard. Screen: You can wipe the screen with a soft cloth to remove any finger marks. Be careful with food and drinks: Avoid eating and keeping glasses of water or cups of coffee near a computer Handle devices carefully: Handle and move your laptop carefully and avoid dropping or banging it against a hard surface. Keep the computer cool: If a computer, laptop or mobile device gets overheated, the internal parts can be damaged. Do not overcharge your battery: Always unplug the device once it is charged 100%. Always plug in devices carefully: Any device being connected to a laptop or computer such as a USB drive or headphones, should be done gently. Do not run too many programs at a time: When too many programs are running at the same time, the computer can become slow and even crash. Backup Your Data Backing up data means to save the information present on your computer on another device Scanning and Cleaning Viruses Sometimes computer viruses can enter a computer from such attacks we can install anti-virus software. Increasing Computer Performance If we have been using a computer for a long time we have a lot of unnecessary files and data, such as temporary files and images. When they use too much hard-disk space, the performance of the computer goes down. It is important that we keep cleaning by removing any extra files. We can use some disk cleaner software, which help us clean up the unnecessary files. Removing SPAM from your Computer Sometimes we get mails from companies who are advertising a product or trying to attract you to their website. Such mails are called SPAM. We should never respond to SPAM and delete it on a regular basis.								
16.	Computer Security Deals with Protecting Computers Computer security and privacy deals with the measures used to prevent loss of data. Reasons for Security Break Security break is leakage of information stored in a computer. Personal information can be lost or leaked in two ways: 1. We are not careful in giving out personal information over the Internet. For example, we share our account details and password on unsecure sites. 2. A person gets unauthorised access to our computer. This can happen in the office if we leave are computer without logging out. Threats to Computer Threats are the ways in which personal information can be leaked from a computer without our knowing. (a) Theft: Theft means stealing of information or hardware. These maybe of three types: • Physical: Where a person may steal your desktop computer or laptop. • Identity: Where a hacker steals your personal information and assumes your identity. Using this false identity, the hacker can gain access to your account information								

	or perform illegal activity. • Software Piracy: This is stealing of software and includes using or distributing unlicensed and unauthorised copies of a computer program or software. (b) Virus: Viruses are computer programs that can damage the data and software programs or steal the information stored on a computer. Major types of viruses are Worms and Trojan Horse. • Worms: These are viruses that replicate themselves and spread to all files once they attack a computer. This makes it very difficult to remove them. • Trojan Horse: A Trojan Horse disguises itself i.e., it appears to be a useful software program but once it reaches a computer it starts behaving like a virus and destroying data. • Online Predator: Online predators are people who trap you into inappropriate relationships. They may be older people posing to be your age, bullying you into doing illegal activities online and sometimes face to face. • Internet Scams: Sometimes you may receive very attractive offers saying you have won a huge money in a lottery and that you can claim the prize by depositing a certain amount of money. When you deposit the money using credit card or online banking, you not only lose the deposit money but your card / account information may be misused later.
17.	Protecting your Data (a) Use passwords to login to your computer (b) Install Anti-virus and Firewall: Anti-viruses and Firewall monitor the data coming in and out of a computer and prevent and viruses from entering. Anti-viruses can also detect and clean viruses that may have entered a computer. (c) Encrypt Data (d) Secure sites: Give details of your credit card or bank account only on secure sites.

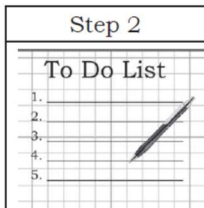
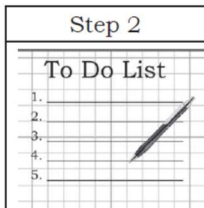
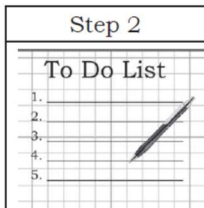
Self-Management Skills

1.	Stress Management refers to focusing human efforts for maintaining a healthy body and mind capable of better withstanding stressful situations. Stress refers to a “physical, mental, or emotional strain or tension”.
2.	Small amounts of stress have a positive effect and they help us stretch ourselves to new level. High levels of stress over prolonged periods need to be managed. A Stress well managed can help one view events and situations as challenges and contributes in the growth of individual. Unmanaged stress leads to anxiety and sorrow resulting in ill status of mental and physical health.
3.	Stress causal agents can have following origins: Mental: Students can be left overwhelmed, if they are unable to handle their assignment submission deadlines and examinations grades. Overly high expectations from self can leave one with chronic anxiety and stress. Physical: As children grow up they may seek more independence, may become critically conscious of their looks and have to cope up with hormonal changes. Issues related to general well-being and health of an individual can lead to low self-esteem and cause stress. Social : Discord amongst family members, peer pressure for doing things which kids will generally avoid , maintaining balance in relationships amongst friends, status show off may lead to stress at times. Financial: Youngsters may have aspirational financial stress. This may of becoming independent or Finances aspirational stress may
4.	Importance of Stress Management: Adequately managed stress prevents medical and physical illnesses. Stress management is vital because it leads to following benefits: Improves mood Boosts immune system Promotes longevity Leads to burst of physical strength, which is vital for goal achievement Complete mental and physical engagement for task accomplishment Increases efficiency and effectiveness Prevents psychological disorders and behavioral problems
5.	Stress Management Techniques Stress is a mental state where one size doesn't fit all, what stresses one person may not stress others at all. It is highly individualistic phenomenon and the solution has to be customized to individual's

	mental state. Signs of Stress: Physical signs which may suggest stress are: breathlessness, dry mouth, butterflies in stomach, indigestion, nausea, acidity, fatigue, sweaty palms, cold hands and feet, irritation and hyperactivity etc. Mental Signs: At the mental level following symptoms may reflect stress irritation, impatience, loneliness, upset, anxious, depressed, frustration, bored, guilty, insecure and forgetfulness. a) Self-awareness amongst individuals for symptoms of stress will help them manage it better. b) They should try understanding the cause, understanding self and taking remedial measures. c) Speaking out to the well-wishers like family and friends is of help in stress management as they provide a social support system. d) Positive thinking e) managing one's time across any or a combination of the following activities aids in stress management. i) <i>Physical Exercise</i>: Physical exercise in the form of walking, skipping or for that matter indulging in any sports has been found to relieve stress as they stabilize mood, improve self-esteem and induce sleep. ii) <i>Yoga</i> iii) <i>Meditation</i> iv) <i>Enjoying</i>: Recreational activities like watching movies, attending concerts, playing games, involving in adventure sports, singing, dancing or even sketching can help individuals transcend to a happier mental state and help manage stress. v) <i>Going On Vacations with Family and Friends</i> vi) <i>Taking Nature Walks</i>: national park or a sanctuary or a trail in country side / village
6.	Ability to Work Independently: Everyone who has been sent on this earth has been blessed in their own way. They together make this earth a beautiful place. Some can sing well, others are good at sports, some are intellectuals and some are good comedians. Everyone is special. But it is important that everyone learns the art of working independently. This typically means that one must learn to take ownership of the task assigned and leave no stone unturned in accomplishing the task. The individual who has the ability to work independently need not be supervised for the task completion.
7.	Importance of The Ability to Work Independently: Ensures greater learning. Individuals feel more empowered and responsible. It provides flexibility to choose and define working hours and working mechanisms. Failure and success of the task assigned are accounted by individuals. Individuals become assets to organizations, groups and nations It ensures creativity and satisfaction amongst individuals.
8.	Enhancing Ability to Work Independently Ability to work independently can be enhanced by being self-aware, self-motivated and self regulated. <i>Self-Awareness</i> Having conscious knowledge of your own self, capabilities, feelings and one's own character is called as self-awareness. It helps individuals to chart their ways for working independently. <i>Self-Motivation</i> Self-motivated individuals have an inner urge to do something, achieve their goals without any external pressure. It is a life skill and must be necessarily developed. <i>Self-Regulation</i> Self-regulation guides independent individuals as it helps them consider long term consequences rather than just transient feelings. It leads to a responsible and a value driven behavior.
9.	Following are some of the skills you must master to succeed in life: Self-awareness: Gather insights on your personality and work-specific proficiencies. Responsibility: Taking responsibility for your tasks is very important. Time Management: Prioritise the things you have to do. Adaptability: Stay current with best practices and read up on new information always.
10.	ABC of stress management A: Adversity or the stressful event B: Beliefs or the way you respond to the event C: Consequences or actions and outcomes of the event

11.	Three Steps to Manage Stress Step 1: Be aware that you are stressed Step 2: Identify what is causing you stress Step 3: Apply stress management methods								
12.	Management Techniques Time management Physical exercise and fresh air Healthy diet Positivity Organising academic life; no delaying Sleep: We should get a good night's sleep for at least 7 hours Holidays with family and friends								
13.	Ability to Work Independently If you can become a calm and relaxed person, you will be have the ability to work independently, which means. 1. becoming self-aware, self-monitoring and self-correcting. 2. knowing what you need to do. 3. taking the initiative rather than being told what to do. 4. recognising your mistakes and not blaming others. 5. having the ability and the will to learn continuously.								
14.	Emotional Intelligence Emotional intelligence is the ability to identify and manage one's own emotions, as well as the emotions of others. It is generally said to include at least three skills: Emotional awareness : the ability to identify and name one's own emotions. • Harnessing emotions : the ability to harness and apply emotions to tasks like thinking and problem solving. • Managing emotions : the ability to regulate one's own emotions when necessary and help others to do the same.								
15.	Some steps to manage emotional intelligence are as given below. Understand your emotions: Observe your behaviour and note the things you need to work on. Rationalise: Do not take decisions abruptly; be rational in your thinking. Practise: Do meditation and yoga to keep yourself calm.								
16.	Self-awareness: Being self-aware means that you can identify your strengths and weaknesses.								
17.	Knowing Yourself: Understanding who you are, what you like or dislike, what are your beliefs, what are your opinions, what is your background, what you do well and what you do not do well is important because only then can you actually measure your strengths and weaknesses								
18.	By understandig ourselves we can • know what we can/cannot do. → Based on that information, we can • plan the next steps to improve ourselves. → By improving, we will • become confident in what we do.								
19.	Difference between Interests and Abilities <table> <tr> <th>Interests</th><th>Abilities</th></tr> <tr> <td>Things that you like to do in your free time that make you happy.</td><td>An acquired or natural capacity</td></tr> <tr> <td>Things you are curious about or would do even if no one asked you to do it.</td><td>Enable you to perform a particular job or task with considerable proficiency.</td></tr> <tr> <td>Things you want to learn or would like to do in the future.</td><td></td></tr> </table>	Interests	Abilities	Things that you like to do in your free time that make you happy.	An acquired or natural capacity	Things you are curious about or would do even if no one asked you to do it.	Enable you to perform a particular job or task with considerable proficiency.	Things you want to learn or would like to do in the future.	
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20.	Self-motivation is simply the force within you that drives you to do things. Self-motivation is what pushes us to achieve our goals, feel happy and improve our quality of life. In other words, it is our ability to do the things that need to be done without someone or something influencing us.								

21.	Types of Motivation Internal Motivation: LOVE We do things because they make us happy, healthy and feel good. For example, when you perform on your annual day function and you learn something new, such as dancing, singing, etc., you feel good. External Motivation: REWARD We do things because they give us respect, recognition and appreciation. For example, Suresh participated in a 100m race and won a prize. This motivated him to go for practice every morning.
22.	Qualities of Self-motivated People  Know what they want from life  Are focussed  Know what is important  Are dedicated to fulfill their dreams
23.	Building Self-motivation Find out your strengths Identify your likes and dislikes. Understand what makes you happy. For example, I love cooking. Set and focus on your goals Define the goals you want to achieve and focus all your energy to achieve your goal. For example, I want to be a chef. Develop a plan to achieve your goals Plan and set timelines to achieve your goals, Plan a list of activities that you will do to achieve each goal. For example, after schooling, you may be required to appear for a competitive examination to join Hotel Management Institute. Stay loyal to your goals Work towards achieving your goal, even when you are facing difficult time. For example, even though I did not clear the Hotel Management entrance exam, I will find out other ways to become a chef.
24.	Goals and Setting SMART Goals Goals : They are a set of dreams with a deadline to get them, for example, saving pocket money to buy a favourite mobile phone by a particular date. Goal setting : It is all about finding and listing your goals and then planning on how to achieve them.
25.	How to Set Goals? We can use SMART method to set goals. SMART stands for: • Specific : A specific and clear goal answers six questions. Who is involved in the goal? What do I want to do? Where do I start? When do I start and finish? Which means do I use? Why am I doing this? Not a specific goal: "I would learn to speak English." Specific goal: "I would learn to speak English fluently by joining coaching classes after my school everyday, and in six months I will take part in the inter-school debate competition." • Measureable : A measureable goal answers the questions "How much?", "How many?" and "How do I know that I have achieved results?" Not measurable goal: "I want to be rich." Measurable goal: "I want to have 5 times more money than what I have today in my hand at the end of this year." • Achievable : Breaking down big goals into smaller parts will make the goal achievable. Bigger Goal: "I want to become a teacher in my school." Breaking it into smaller goals: . Complete higher secondary . Complete Graduation . Complete B.Ed. . Apply for jobs in the teaching field • Realistic : A realistic goal would be something that we want to achieve and can work towards. Example of unrealistic goal: "I will read my entire year's syllabus in one day and get good marks."

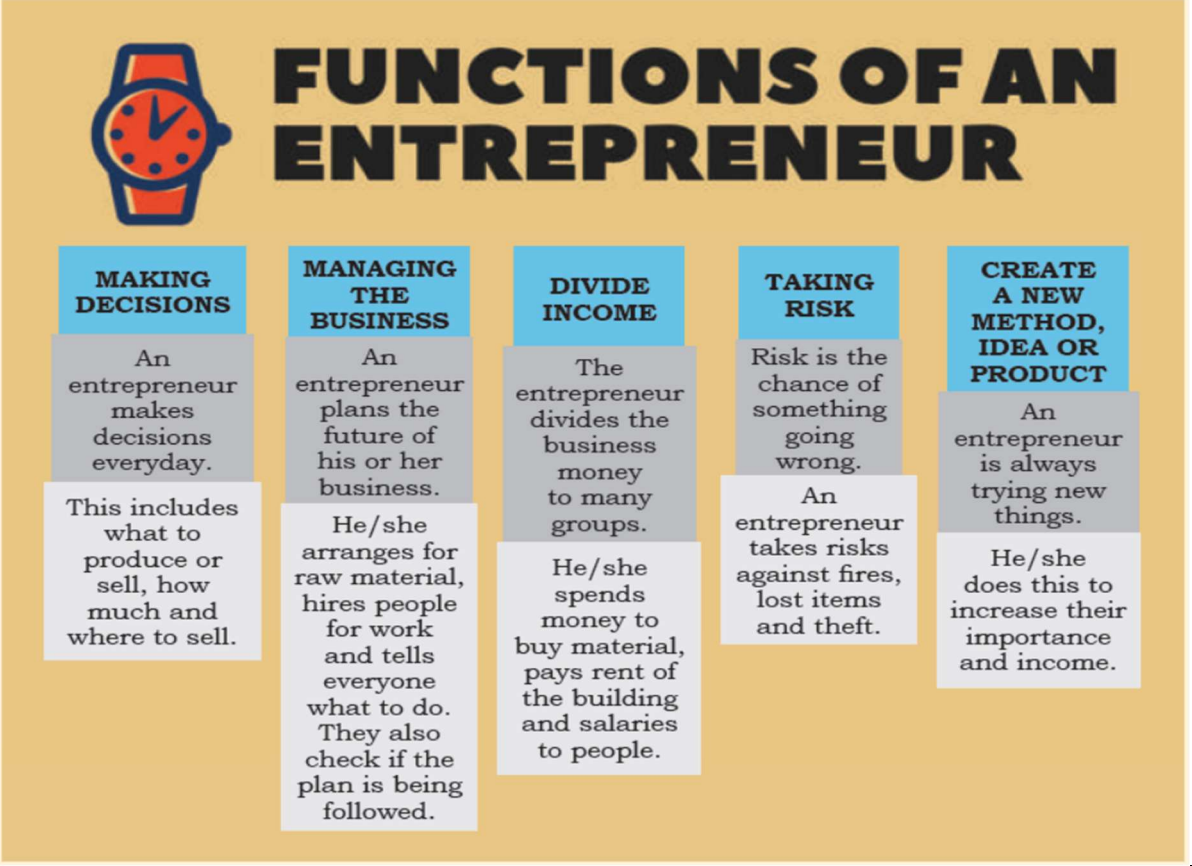
	Realistic goal: "I spend 3 hours every day of the year after school to revise my subjects to get good marks in the exams." • Time bound : A SMART goal should have a timeframe by when the goal needs to be achieved. This encourages us to take actions to completely fulfill the goals. Not a time bound goal: "I want to lose 10 kg someday." Time bound goal: "I want to lose 10kg in the next 6 months."								
26.	Self-regulation — Time Management Time management is the ability to plan and control how you spend the hours of your day well and do all that you want to do. An example of good time management skills would be when you decide to finish your homework immediately after school so you have time to watch TV later in the evening.								
27.	Time management is the thinking skill that helps you to • complete tasks on time. • make a daily timetable. • make a good guess at how long it will take you to do something. • submit homework and assignments on time. • not waste time during the day.								
28.	Four Steps for Effective Time Management <table><tr><td> Step 1 </td><td> Step 2 </td><td> Step 3 </td><td> Step 4 </td></tr><tr><td> Organise • We plan our day-to-day activities. • We make a timetable that we follow. • We keep our surroundings and study table clean and mess free. • We put things back where they belong. • It helps us save time! </td><td> Prioritise • We make a to-do list that has all our activities and we rank them in the order of importance. For example, you may rank doing homework as the most important task. It helps us to get the most important task done first and also to track what is pending. </td><td> Control • We have a control over our activities and time. • We avoid time wasters like chatting on the phone, surfing gossip sites, etc., and focus on more important things. </td><td> Track • We identify and note where we have spent our time. This will help us analyse if we have used our time effectively or not. It also helps us to identify time-wasting activities </td></tr></table>	Step 1 	Step 2 	Step 3 	Step 4 	Organise • We plan our day-to-day activities. • We make a timetable that we follow. • We keep our surroundings and study table clean and mess free. • We put things back where they belong. • It helps us save time!	Prioritise • We make a to-do list that has all our activities and we rank them in the order of importance. For example, you may rank doing homework as the most important task. It helps us to get the most important task done first and also to track what is pending.	Control • We have a control over our activities and time. • We avoid time wasters like chatting on the phone, surfing gossip sites, etc., and focus on more important things.	Track • We identify and note where we have spent our time. This will help us analyse if we have used our time effectively or not. It also helps us to identify time-wasting activities
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29.	Tips for Practicing the Four Steps for Effective Time Management • Avoid delay or postponing any planned activity • Organise your room and school desk • Develop a 'NO DISTURBANCE ZONE', where you can sit and complete important tasks • Use waiting time productively • Prepare a 'To-do' list • Prioritise • Replace useless activities with productive activities								

Entrepreneurial Skills

1.	While the concept of entrepreneurship has a fairly wide meaning, it is simply described as starting a business using the resources available to a person. An entrepreneur combines factors in a creative manner in order to generate value for customers and create wealth.
2.	The beauty of entrepreneurship in a society is that as businesses prosper the impact the businesses create also increases. It is due to entrepreneurial activity that the society is provided with goods and services. In turn, society provides market for products and services provided by the entrepreneur.
3.	Entrepreneurship has some positive impact on society such as: • Accentuates economic Growth • Fosters Creativity

	• Stimulates Innovation and Efficiency • Creates Jobs and Employment Opportunities • Solves the problems of the society • Encourages welfare of the society
4.	Society plays a role in boosting entrepreneurship by: • Creates needs and demands • Provides raw materials • Enables financial support • Creates a need for education • Catalyses policy formation and reform • Facilitates networking • Supports infrastructure development However, entrepreneurship also creates some adverse impact on society sometimes. This happens in the form of environmental degradation, trade imbalance, labour exploitation etc.
5.	Some common entrepreneurship activities related to society include: Social Entrepreneurship: Social entrepreneurship can be understood as creation of sustainable solutions for social problems that leads to social change by employing entrepreneurial mindset, processes and operations. Agricultural Entrepreneurship: Agricultural Entrepreneurship can be defined as being primarily related to the marketing and production of inputs and products used in agricultural activities. Women Entrepreneurship: Women entrepreneurship is referred to the entrepreneurial activity led by women, where women undertake risks, create enterprises, organise factors of production, innovate with products/services and generate employment opportunities. Small Scale Entrepreneurship: Small scale entrepreneurship refers to starting industries in which manufacturing, trading, providing services, productions are done on a small scale or micro scale.
6.	Qualities of an Entrepreneur Hard work: Without working hard, no entrepreneur can be successful. Optimism: Positivity and belief in what they do is what takes entrepreneurs far in their journey. Independence: Entrepreneurs are confident and like the freedom to take decisions. Energetic: Drive and energy is always high in successful entrepreneurs which makes them extremely proactive. Self-confident: Belief in one's own vision and abilities and passion for the goal makes successful entrepreneurs confident. Perseverant: A way of thinking that helps to accept failure, learn from mistakes, not give up, continuously experiment etc. is crucial for an entrepreneur.
7.	Functions of an Entrepreneur Organisation Building and Management: Organisation building is about bringing together different factors of production and allocating these resources to bring down costs and losses. Risk taking: Risk taking is about taking responsibility and planning for a loss or mishap that may occur in the future due to unforeseen contingencies. Innovation: Entrepreneurs innovate by introducing new concepts, products, services, designs, ideas etc.
8.	Promotional functions Idea Discovery: The first step towards entrepreneurship is discovery of a business idea. Detailed Investigation: While coming up with ideas can seem like an easy process, the challenge is to understand if the idea has the potential to turn into a viable business venture. Assembling the Requirements: No matter what is the type of the business, an entrepreneur has to work towards assembling and organizing all the requirements once he/she is sure about the practicality and profitability of the proposition. Financing: Raising capital for a business is one of the core functions that entrepreneurs perform themselves, not just at the initial stage of the business, but even to grow and scale the business.
9.	Managerial functions

	Planning: An entrepreneur documents a business idea in the form of a business plan, to detail each element of the business such as product or service description, operations, marketing, finance, accounting, growth plan etc. Organizing: Organizing in the managerial context refers to setting specific and attainable goals and objectives to be achieved by different departments and by each employee. Directing: Directing is about initiating planned action and ensuring that each employee is performing effectively. Staffing: Staffing refers to different sub-processes around human resource management, such as manpower planning, recruitment, selection, placement, training, transfer, salary, promotion, payroll, appraisal etc. Leadership: Leadership is more of a skill than a function for an entrepreneur as he or she has to lead, guide and supervise people who work for the business. Communication: Communication enables exchange of feelings, ideas, emotions, information and knowledge between two or more persons Supervision: Supervision is a key skill as even overseeing that things are being executed well is important. Motivation: Along with guiding employees, an entrepreneur also plays a role of motivating them and encouraging positive behaviour and culture within the organization. Co-ordination: Co-ordination between different departments of the business helps to ensure that work happens in a timely manner. Controlling: Sometimes controlling is necessary for the entrepreneur to set company rules and policies. Negotiation: An entrepreneur negotiates terms and conditions for many parts of the business such as payment terms and timelines, salaries of employees, vendor quotations etc.
10.	Commercial Functions Production and Operations Finance and Accounting Marketing Human Resource Management
11.	Role of Entrepreneurs Innovator's Role Agent's role Coordinating role Risk assumption role Capital formation role Imitating role Employment Generation role Status transformation role Balancing role
12	Importance of Entrepreneurs Free market evolution: In a free market, entrepreneurs are able to respond to changing customer preferences, prices, shifts in demand etc. Efficiency improvements: Entrepreneurs tend to adapt in an established business and increase their own efficiency. New markets: Entrepreneurs can often 'redefine the rules' of an established industry. New values: Sometimes, entrepreneurs choose ethics over profit and offer a more ethical product to the world and are transparent about it.
13	Myths of Entrepreneurship Entrepreneurs are born a certain way Great ideas are what makes entrepreneurs Entrepreneurs have to take a lot of risk Businesses either skyrocket or fail A lot of money is required to start any business One must know everything before starting a business
14	Entrepreneurship as a Career – Why Entrepreneurship for You?

	Nurtures development of entrepreneurial skills and capabilities: Entrepreneurship cultivates unique skills and encourages outside the box thinking. Enables application of an entrepreneurial mindset: Entrepreneurial mindset and skills are not necessarily only relevant to becoming an entrepreneur, but also in any field. Develops the ability to handle failure and ambiguity Enhances critical thinking and problem-solving ability Provides early exposure to the real world Inspires to think about one's career differently Leads to creating difference in society: Entrepreneurs not only economically impact society but they also work on identifying the real needs and problems of people and solving them.
15.	Difference between Entrepreneur and Business Ramya and Ramu both own plant shops. Ramu sits at his shop every day. When customers come, he sells to them. Ramya walks around and gets customers to her shop. She also sells seeds and flowers with her plants. Ramya is working to do more and grow her business. She has also added a different idea to her business. So, Ramya is an entrepreneur and Ramu is a businessman.
16.	what do entrepreneurs do when they run their business? Fulfil Customer Needs Use Local Materials Help Society Create Jobs Sharing of Wealth Lower Price of Products
17.	Qualities of an Entrepreneur 1) CONFIDENT 2) Keep TRYING NEW IDEAS. 3) PATIENT 4) CREATIVE 5) Take RESPONSIBILITY for their mistakes 6) take DECISIONS AFTER THINKING about whether they are good or bad for my business. 7) WORK HARD 8) DO NOT GIVE UP.
18.	 FUNCTIONS OF AN ENTREPRENEUR MAKING DECISIONS An entrepreneur makes decisions everyday. This includes what to produce or sell, how much and where to sell. MANAGING THE BUSINESS An entrepreneur plans the future of his or her business. He/she arranges for raw material, hires people for work and tells everyone what to do. They also check if the plan is being followed. DIVIDE INCOME The entrepreneur divides the business money to many groups. He/she spends money to buy material, pays rent of the building and salaries to people. TAKING RISK Risk is the chance of something going wrong. An entrepreneur takes risks against fires, lost items and theft. CREATE A NEW METHOD, IDEA OR PRODUCT An entrepreneur is always trying new things. He/she does this to increase their importance and income.
19.	Myths about Entrepreneurship

	The misconception is that every business idea needs to be unique or special. The misconception we have is that a person needs a lot of money to start a business. A misconception we have is that only a person having a big business is an entrepreneur A misconception we have is that entrepreneurs are born, not made.												
20.	Entrepreneurship as a Career Option A career is a line of work that a person takes for life. There are two ways a person can earn a living. They are 1. self-employment 2. wage employment For example, a doctor who works for a hospital is a wage employed person. If the same doctor has their own clinic, he or she is a self-employed person. A self-employed person running a business to satisfy a need by trying new ideas is an entrepreneur.												
21.	A person who becomes an entrepreneur goes through a career process. <table><tr><th>ENTER</th><th>SURVIVE</th><th>GROW</th></tr><tr><td></td><td></td><td></td></tr><tr><td>When an entrepreneur is starting, they are just entering the market to do business.</td><td>There are many entrepreneurs in the market. The entrepreneur has to remain in a competitive market.</td><td>Once the business is stable, an entrepreneur thinks about expanding his or her business.</td></tr><tr><td>For example, Sanjana is starting a small grocery store in a locality.</td><td>For example, there are many other grocery stores in the area. Yet, Sanjana's store survives the competition and does well. She also expands the store to two more floors.</td><td>For example, after five years, Sanjana has opened a chain of four more grocery stores in the same city. In the next two years, she plans to expand to tow other cities.</td></tr></table>	ENTER	SURVIVE	GROW				When an entrepreneur is starting, they are just entering the market to do business.	There are many entrepreneurs in the market. The entrepreneur has to remain in a competitive market.	Once the business is stable, an entrepreneur thinks about expanding his or her business.	For example, Sanjana is starting a small grocery store in a locality.	For example, there are many other grocery stores in the area. Yet, Sanjana's store survives the competition and does well. She also expands the store to two more floors.	For example, after five years, Sanjana has opened a chain of four more grocery stores in the same city. In the next two years, she plans to expand to tow other cities.
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Green Skills

1.	Over the years, with economic development, there has been an increase in environmental pollution. For example, with the introduction of high input agriculture, we can grow more food by using fertilisers, pesticides and hybrid crops. But it has led to soil and environmental degradation. We need to plan the use of resources in a sustainable manner so that we and our future generations can enjoy the good environment
2.	Sustainable Development: Sustainable development is the development that satisfies the needs of the present without compromising the capacity of future generations, guaranteeing the balance between economic growth, care for the environment and social well-being.
3.	Importance of Sustainable Development: With increasing population and income, the consumption of goods is increasing day by day. This has led to increase in production and utilisation of natural resources, which are required for producing goods. Society must thus change its development strategy to a new form where development will not destroy the environment. This form of sustainable development can only be brought about if each individual practices a sustainable lifestyle. Since most of the natural resources are scarce, therefore, careful utilisation of resource is necessary.
4.	Problems Related to Sustainable Development: Three major problems related to sustainable development are: (a) Food: The amount of rich, fertile land needed to grow crops, such as wheat, rice, etc., is becoming less as we are using up more and more land for other purposes. Soil nutrients are also getting depleted and lots of chemicals are spoiling the soil due to use of chemical fertilisers. (b) Water: We use fresh water from rivers and ponds for drinking and cleaning but dump garbage into them. The rivers and ponds are getting polluted. This way after several years, we will have no clean water for our use. (c) Fuel: We are using a lot of wood from trees as fuels and for construction of homes and furniture. As more and more trees are being cut, it is affecting the climate of the place. Extreme weather conditions, such as floods, extreme cold or heat, are seen in many places, which affect the people living there.

5.	An environment friendly way for production of power is using solar energy or energy from the sun which is unlimited. One major example of large scale production of solar power is seen in Charanka — Gujarat Solar Park. This barren land with wild vegetation has a mega solar power plant of 600 MW. This has reduced the dependence on non-renewable fossil fuels in Gujarat. The people of Charanka have benefited as they have a good source of income and this will also help future generations in the next 40–50 years to come.
6.	Sustainable development includes • reducing excessive use of resources and enhancing resource conservation; • recycling and reuse of waste materials; • scientific management of renewable resources, especially bio-resources; • planting more trees; • green grassy patches and trees to be interspersed between concrete buildings; • using more environment friendly material or biodegradable material and • use of technologies, which are environmental friendly and based on efficient use of resources.
7.	Sustainable Development Goals: The Sustainable Development Goals (SDGs) are a universal call of action to end poverty, protect the planet and ensure that all people enjoy peace and prosperity. The Sustainable Development Goals (SDGs) were launched at the United Nations Sustainable Development Summit in New York in September 2015, forming the 2030 Agenda for Sustainable Development. It has set targets that the countries have to should work towards and achieve by 2030.
8.	The 17 SDGs have been made with the aim to take care of important issues facing businesses, governments and society. Some of these issues are poverty, gender equality, water use, energy, climate change and biodiversity. Countries are now making policies and regulations that will promote sustainable systems needed in all economic sectors to provide a secure, affordable and sustainable economy.
9.	Sustainable Development Initiatives When plastics were banned by the municipal corporation of Mangalore, an innovator and entrepreneur Mr Avasth Hedge found an eco-friendly alternative. He made a 100 per cent bio-degradable bag which can dissolve in hot water and decompose in natural environment. This is now being used in many countries and is going to help the environment. Another innovative idea was to stop the use of plastics spoons, forks, etc. Narayana Peesapaty made edible cutlery made out of a grain — these spoons can be eaten and even if they are not eaten they will easily decompose in the soil enriching it. <https://www.ibef.org/Innovations-from-India.aspx>
10.	Sustainable Processes Some practices, such as organic farming, vermi-composting and rainwater harvesting are being used to help preserve the environment. Organic farming is where farmers do not use chemical pesticides and fertilisers to increase their production. They use organic and natural fertilisers, such as cow dung to help in growing crops. This helps in better quality chemical free crops while at the same time maintaining the soil quality for future use. This is a true example of sustainable development where we are not only using the earth resources but are also preserving it for our future generations.
11.	Our Role in Sustainable Development Increasing population and development has led to the increase in consumption of the natural resources. The more the population, the more food, energy and water we need. When we grow more crops, the soil's nutrients are consumed and slowly the soil becomes unusable. Similarly, if we continue to use fossil fuels, such as coal, oil and natural gas, very soon we will run out of these natural resources. We use so many resources from nature but whatdo we give back to nature? Natural resources are limited and with time they will get over and if we do not do anything about it, our future generations will not be able to survive. One initiative was taken by Ministry of Railways in Varanasi and Raibareilly where they introduced clay pots '<i>Kulhads</i>' to replace plastic and paper cups and to bring back taste of tradition. Three major advantages gained by using '<i>Kulhads</i>' are to 1. reduce cutting of tree for making paper cups. 2. job creation for potters, which contributes to economy. 3. reduce plastic waste.
12.	Our Role towards Sustainable Development United Nations has formed 17 Sustainable Development Goals (SDGs) so that countries and nations can work towards finding a solution to the growing problems in the world. Countries should form rules and guidelines to remove poverty and hunger and provide good health and quality education. These goals have

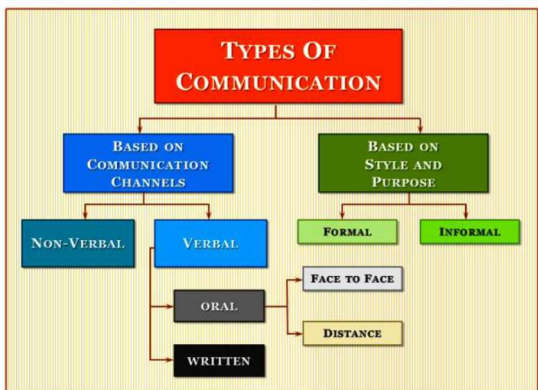
	been made so that all the nations take action to protect our air, water and land from pollution and provide clean water, air and energy to all people. Sustainable development can actually happen only when each one of us works towards it. We have to become responsible environment citizens who can protect the environment through our own efforts.
13.	Here are some basic ways in which people can help towards Sustainable Development Goals. Quality Education: Education is the most important factors for sustainable development. Children who have gone to school will be able to do jobs so that they can take care of themselves and their families. Education helps us become aware of our role as a responsible citizen. We should 1. use the facilities present in our areas. 2. take our friends to school. 3. help friends study. 4. stop friends from dropping out of school. Clean Water and Sanitation We must make efforts to make India free of open defecation by building toilets and creating awareness towards sanitation. Industrial pollution is polluting our water resources, which in near future will cause scarcity of clean drinking and usable water. We must take measurable steps by promoting awareness to keep water sources clean. Affordable and Clean Energy Using solar power i.e., power generated using the sun does not cause pollution as it does not require burning of non-renewable fuels, such as coal. We are making efforts to increase the solar power generation so that our electricity needs are met and at the same time we do not pollute the environment or use up natural resources. Use of biogas is also an eco-friendly alternative to natural gas. Decent Work and Economic Growth We can (a) study and find good jobs to take care of ourselves and our families. (b) work hard and contribute to society. (c) learn and develop skills so that we get add value in our community. Reduced Inequalities To reduce inequalities we can 1. be helpful to one another. 2. be friendly with everyone. 3. include everyone while working or playing. 4. help others by including everyone whether they are small or big, girl or boy, belong to any class or caste. Sustainable Cities and Communities Creating Sustainable Cities 1. Save energy by switching off lights and fans when not in use. 2. Use natural light as much as possible. 3. Use energy efficient lights (LED bulbs) and appliances. Responsible Consumers and Producers We can become responsible about our own environment by 1. reusing paper, glass, plastic, water, etc. 2. taking cloth bags to market carrying fruits and vegetables. 3. donate things we do not use such as clothes, books, furniture, food, etc. 4. Buy and eat seasonal fruits and vegetables from local growers. 5. Repair leaking taps and pipes to avoid wasting water. 6. Sort and treat garbage before disposing. Protect Life Below Water Tons of plastic is found in the seas, which is killing marine life. Protecting marine life, saving our oceans from pollution is necessary to preserve the marine life. Protect Life on Land Cutting of trees is leading to soil erosion and making land dry and unusable for cultivation. Planting more tree to replace the ones that we have cut is an important step towards sustainable development.
14.	In our daily life we can contribute to create a Sustainable Society by following 4Rs' and 1U of Sustainability. These are REFUSE, REDUCE, REUSE, RECYCLE and UPCYCLE. To create a sustainable environment, we must first REFUSE to use products that may harm the environment. REDUCE is the next step, that is minimizing the use of the products that may cause harm to environment.

We must **REUSE** these products as far as possible, in order to reduce the waste generation. Next comes **RECYCLE**. After reusing the product, we must try to recycle it as far as possible. When we recycle any product, the quality is usually affected and the new product is not of the same quality as the original product.

In these modern times, another smart way has seen its emergence, this is the U in sustainability – **UPCYCLING**. This allows the scope of human skills to get their due. Any product that is not usable can be upcycled with creativity and innovation. UPCYCLING of products could be done manually as well as with help of machines, giving a new look to the old product and making it look desirable.

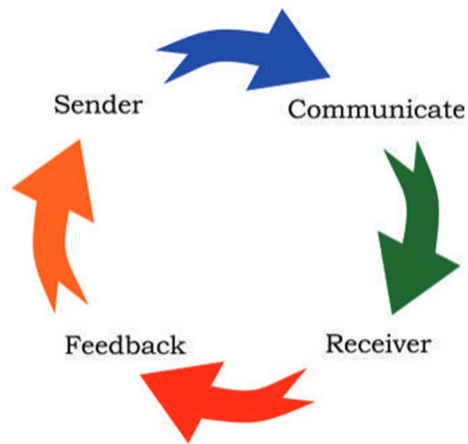
By following the 4 R's and 1 U, we all can contribute in reducing waste generation thereby protecting the resources and keeping the environment safe.

Communication Skills

1.	Communication is the imparting or exchanging of information by speaking, writing or using some other medium and means of sending or receiving information.
2.	
3.	Learning objectives of Effective communication 1. Development of Interpersonal Skills 2. To express effectively & with maximum efficiency
4.	Writing is a form of communication that allows students to put their feelings and ideas on paper, to organize their knowledge and beliefs into convincing arguments, and to convey meaning through well-constructed text. In its most advanced form, written expression can be as vivid as a work of art.
5.	Phrases - Phrases are a group of words that work together to communicate an element of speech.
6.	Sentences : A set of words that is complete in itself, typically containing a subject and predicate, conveying a statement, question, exclamation, or command, and consisting of a main clause and sometimes one or more subordinate clauses.
7.	4 Types of Sentences Declarative Sentence • Tells something. • Ends with a period. (.) Interrogative Sentence • Asks a question. • Ends with a question mark. (?) Exclamatory Sentence • Shows strong feeling. • Ends with an exclamation mark. (!) Imperative Sentence • Gives a command. • Ends with a period or an exclamation mark. (. or !)
8.	Parts of speech - A category to which a word is assigned in accordance with its syntactic functions. In English the main parts of speech are noun, pronoun, adjective, determiner, verb, adverb, preposition, conjunction, and interjection.

9.	Article writing is the process of creating a non-fiction text about current or recent news, items of general interest or specific topics.
10.	A paragraph is a series of sentences that are organized and coherent, and are all related to a single topic.
11.	Barriers to Effective Communication A communication barrier is thus anything that prevents us from receiving and understanding the messages others use to convey their information, ideas and thoughts. Physical Barriers – Physical barriers separate people from each other and mark territories. Language Barriers – Not using words another can understand will certainly stop your message from being conveyed. Gender Barriers – Variation exists among masculine and feminine styles of communication. Attitudinal Barriers - as those behaviors or perceptions that are divisive in nature – the ones that can lead to nagging doubt, sullen disagreement or even overt conflict. Perceptual Barriers – Different world views can create misunderstanding. People tend to interpret messages from their own point of view or ideologies. Cultural Barriers – Ethnic, religious, and social differences can often create misunderstandings when trying to communicate. Emotional Barriers –If one is consumed with emotion he will have difficulty in understanding what is communicated. Answer the following quiz questions- - Not stating instructions clearly is an example of..... (language barrier) - Not understanding the customs or traditions of a speaker could mean there is a..... (cultural barrier) - When you are overcome by your own feelings and unable to communicate well it means there is an(emotional barrier) - Using slang or words that are used within your own social group is an example of a (cultural barrier) - You may like to have your own private cabin to work in but if you are not willing to meet your team face to face it is an indication of a..... (physical barrier) - You have an..... (attitudinal barrier) if you think you are more qualified than others and are superior in knowledge. - If you have a totally different viewpoint from your team mates on the way a training programme is to be planned you have a(conceptual barrier)
12.	Handling Barriers to Communication Engage in face-to-face interactions whenever possible. Reduce or eliminate distractions around you. Try to assess the needs of the receiver. Hone your active listening skills. Soften your tone and language. Encourage feedback.
13.	Described below are some common barriers to effective communication which we must avoid: Second guessing the sender: We do this when we are impatient with the speaker and are in a hurry to finish the sentence for the speaker. Stereotyping: We often form stereotypes about those whom we know the least! Once our mental sets are created, all our transactions are affected by these sets, preventing us from effective listening. Halo effect: This is another form of stereotyping. Based on a single characteristic we make up our mind usually positively about the other person. Not listening as a status or gender issue: Studies have shown that men listen much less than women do. Similarly those in positions listen less to those who are lower in hierarchy.
14.	Communication has three important parts: Transmitting — The sender transmits the message through one medium or another. Listening — The receiver listens or understands the message. Feedback — The receiver conveys their understanding of the message to the sender in the form of feedback to complete the communication cycle. The various elements of a communication cycle are: Sender: the person beginning the communication.

	Message: the information that the sender wants to convey. Channel: the means by which the information is sent. Receiver: the person to whom the message is sent. Feedback: the receiver's acknowledgement and response to the message. Methods of Communication: Face-to-face informal communication e-mail Notices/Posters Business Meetings Other Methods
15.	Verbal Communication: Verbal communication includes sounds, words, language, and speech. Speaking is one of the most effective and commonly used way of communicating. Type of Verbal Communication: Interpersonal Communication Written Communication Small Group Communication Public Communication Advantages of Verbal Communication It is an easy mode of communication in which you can exchange ideas by saying what you want and get a quick response. Verbal communication also enables you to keep changing your interaction as per the other person's response. Disadvantages of Verbal Communication Since verbal communication depends on written or spoken words, sometimes the meanings can be confusing and difficult to understand if the right words are not used. Mastering Verbal Communication Think Before You Speak Concise and Clear Confidence and Body Language
16.	Non Verbal Communication Non-verbal communication is the expression or exchange of information or messages without using any spoken or written word. Importance of Non-verbal Communication In our day-to-day communication • 55% communication is done using body movements, face, arms, etc. • 38% communication is done using voice, tone, pauses, etc. • only 7% communication is done using words. Types of Non-verbal Communication Gestures or Body Language Posture Facial Expressions Touch Space Eye Contact Paralanguage Visual Communication: Visual communication proves to be effective since it involves interchanging messages only through images or pictures and therefore, you do not need to know any particular language for understanding it.
17.	Communication Cycle and Importance of Feedback



Feedback can be positive or negative.

A good feedback is one that is:

- specific
- timely
- Polite
- Offering continuing support

Types of Feedback

Positive Feedback

Negative Feedback

No Feedback : It is also a feedback in itself which indicates disagreement of ideas.

Importance of Feedback

- It validates effective listening
- It motivates
- It is always there
- It boosts learning
- It improves performance

18. 7 Cs i.e., Clear, Concise, Concrete, Correct, Coherent, Complete and Courteous.
Absence of any of these 7Cs can lead to miscommunication.

